

Artistic Exploration of a Pixilated Choreography in Virtual Reality: Exploring 3D Gaussian Splatting for Stop-Motion Animation

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ABSTRACT

Part four of the project *VRinMotion* explores the integration of stop-motion pixilation in virtual reality environments, using live actors as frame-by-frame subjects. In collaboration with artists Paul Wenninger and Raul Maia, the project employs a unique workflow combining 35 synchronously triggered cameras and 3D Gaussian splatting to capture volumetric animation. This approach preserves the tactile aesthetics of stop motion while enhancing the narrative potential of both mediums. The project investigates new visual and spatial dynamics, such as mixing scales between miniature environments and full-scale human actors, and explores the creative possibilities for storytelling in virtual spaces.

Keywords: Virtual Reality, Stop-Motion Animation, Pixilation, Art-Based Research, Gaussian Splatting

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Applied computing—Arts and humanities—Media arts

1 INTRODUCTION

In its fourth and final segment of *VRinMotion*, *ExperiMotion 4*, the art-based research project highlights the integration of the classic stop-motion technique, pixilation, into virtual reality (VR) environments. Instead of inanimate objects, live actors are used as frame-by-frame subjects. At the heart of this work-in-progress is a collaboration with invited guest artists Paul Wenninger and Raul Maia.

The integration of volumetric pixilation animation in VR is achieved through a workflow involving an animation setup with 35 synchronously triggered cameras and a 3D reconstruction method called 3D Gaussian splatting. Although there have been previous projects dealing mainly with VR puppetry and 3D-capturing miniature characters, there are some striking differences in the current approach. Apart from the obvious size differences in the recorded characters, the aim is not to create a manipulable, three-dimensional character model, but rather to capture, generate, and assemble a sequence of individual 3D snapshots from a performance.

The desired result seamlessly transfers the tactile aesthetics of pixilation into the immersive and interactive spaces of VR, expand-

ing the expressive and narrative potential of both mediums. Beyond the inherent affordances of virtual reality, such as unrestricted movement, numerous less evident opportunities arise. For instance, the ability to mix scales through 3D capturing backgrounds or environments at a miniature level while capturing human actors and objects at actual scale, as is common in pixilation. Aspects like these not only generate interesting new visual aesthetics in VR but also open up possibilities for exploring the perceptual dynamics of spatial relationships in immersive storytelling. In the project's art-based research process, the following questions were developed:

1. How can pixilation (i.e. stop motion using humans as frame-by-frame subjects) be translated into a virtual environment?
2. How does the stop-motion setup and workflow differ from classical camera setups and surroundings during the recording process?
3. What are the advantages and disadvantages of using the radiance-field method of Gaussian splatting to transform 2D multiview footage into 3D representations?
4. Which useful workflows can be developed to compose a scene from different elements (protagonist, objects, models, and environments/surroundings)?
5. What dramaturgical/choreographical possibilities or restrictions emerge for the artistic decisions? How can the viewer's gaze be guided in a looped virtual environment?

2 THEORY

While pixilation is an established and well-documented animation technique, incorporating pixilation animation into VR is still a novel approach. The integration of pixilated performances into VR is achieved through a workflow involving Gaussian splatting.

2.1 Pixilation

In the analog animation technique pixilation actors adopt a sequence of poses, each pose captured as a single frame, then adjust their positions incrementally for subsequent captures. The term was invented by the filmmakers Norman McLaren and Grant Munro for the film *Neighbours* [18], and further explored in *Two Bagatelles* [20] as well as the experimental film *Pas de Deux* [19] (see [5], [6], [21, p. 110]) or the work of Jan Svankmajer (see [36]). The technique itself can be traced back to frame-by-frame manipulation of live human actors by film pioneers such as Georges Méliès (see [8], [17]). It even reaches beyond the beginning of film history when artistic scientists such as Eadweard Muybridge and Étienne-Jules Marey experimented with pre-cinematic motion studies and chronophotography in the 19th century (see [22], [7]). However, pixilation works exactly in the

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opposite way of chronophotography. Whereas the latter broke down continuous movement into individual still images, pixilation creates continuous movement from individual still moments. Although it is an old analog animation technique, it is used for contemporary experimental music videos and by artistic filmmakers such as Juan Pablo Zaramella and his short film *Luminaris* [35] (see [9]) or Daria Kashcheeva in the pixilation live-action hybrid *Electra* [13] (see [24])

2.2 Artistic Input

ExperiMotion 4 was shaped by the previous work of guest artist Paul Wenninger and his experience as a dancer, choreographer, and filmmaker with a focus on pixilation. The animation technique is extremely challenging for the actors and the recording process. After all, the actor has to hold the pose that corresponds to the next phase of his movement until the set and camera position have been adjusted according to the director's wishes. As Paul Wenninger explained in an interview [33], he usually works with a seven-axis camera crane to control movement when he wants to incorporate camera movement into the shot in addition to the actor's movement. As a result, pixilation shoots are quite long: about a week's work of animation for 20 seconds of film or video.

In *Trespass* [30] the pixilated filmmaker himself jumps back and forth from an enclosed room to various locations in Eritrea, Brussels, Helsinki, Vienna, and France while continuing his jerky movements. In *O* [32] Wenninger remains still, but nevertheless animated, in the middle of the frame. An unrenovated room shifts around him and through the large windows time-lapse projections of the outside are visible. Raul Maia, a physical theater artist, who worked on several of Wenninger's films as an actor, also joined the project. In *Uncanny Valley* [31] Maia, Wenninger and Jan Jakubal portray soldiers, animated through pixilation, struggling within found footage material from World War I. For *Dead Reckoning* [34], which Wenninger realized in collaboration with Susan Young, Maia travels in highspeed through the inner city of Vienna.

2.3 3D Stop Motion in VR

Several initiatives have already explored the integration of 3D stop motion-style animations into virtual reality. For example, software tools such as *Tvori* [29] facilitate character animation in a manner similar to traditional digital content creation packages such as *Blender* or *Maya*. Using stepped keyframes, these tools can replicate distinctive motion characteristics of stop-motion animation (see [12]). However, without additional tools for altering textures and/or geometry dynamically, these methods typically lack the tangible qualities that define traditional stop-motion techniques. Therefore, efforts to capture and incorporate stop-motion animated objects into VR have been explored in several projects: *Near-Field VR* [2] by the University of Southern California combines light-field rendering with stop motion. The *Stop-Motion VR* [15] project at the Konrad Wolf Film University of Babelsberg employs photogrammetry to bring physical stop-motion objects into VR. In the unreleased adventure game *The Midnight Walk* [11], handcrafted clay miniatures were meticulously 3D-captured and brought to life in a stop-motion-inspired animation style. *Mad God* [28] by Phil Tippet, *Gymnasia* [16] by Chris Lavis and Maciek Szczerbowski, and *The Sick Rose* [27] by Zhi-Zhong Tang and Yun-Hsien Huang have utilized 360-degree (stereoscopic) cameras to create immersive environments featuring stop-motion animation.

For immersive pixilation, an example is the pixilation recorded by a fisheye camera used at Robert Morris University to introduce students to immersive Fulldome applications. Furthermore, in a previous phase of the project, *ExperiMotion 2*, a pixilation-inspired method has already been used, working with the rgb and depth data captured by an Azure Kinect (see [23]). However, this special

application of pixilation in VR being explored in *ExperiMotion 4*, has not yet been investigated in any previous project.

2.4 Photogrammetry und Gaussian Splatting

To achieve volumetric reconstruction of real objects or spaces for VR rendering it is common to apply photogrammetry-based methods. In those, a set of photos of the same subject from different perspectives is used to triangulate a textured mesh representation of the volume of interest. However, these methods usually don't work well in object reconstruction with specular and transparent surfaces, as well as fine details structures, such as hair. Only recently several methods have emerged that have made the capturing and rendering of radiance fields (i.e. the light transport in space at any point) much more accessible. Particularly the method of 3D Gaussian splatting (3DGS) [14] and derivatives of it get lots of attention, as they allow for photorealistic volumetric reconstruction of real environments. Instead of textured meshes, the output is a type of improved point cloud, where each point is rendered as a 3D Gaussian splat with view dependent color response (based on spherical harmonics). A single scene can consist of up to a million individual splats, which form a volumetric representation like grains. They can still be rendered at interactive framerates on modern GPUs, even for VR. The process to create a single capture involves the multiview-stereo capturing of the scene (i.e. photos from many different angles), a subsequent pose estimation of the cameras/images and then an optimization process for the creation of the 3DGS representation.

Even though the method is relatively new, several software frameworks exist (e.g., [26]) that allow for the creation of the splats. Therefore, the project team decided to use this method for *ExperiMotion 4*, as the visual fidelity of the captured scenes have a visual style preferable over classical photogrammetry, where the mesh representation tends to be imperfect and visually displeasing.

3 METHODS AND OUTCOME

Like previous segments of *VRinMotion*, *ExperiMotion 4* is based on an art-based research approach. It aligns with Borgdorff's threefold characterization of artistic research: Research on the Arts, Research for the Arts, and Research in the Arts [3, p. 76]. *VRinMotion* navigates this spectrum by engaging both theoretical and practical aspects, where artistic practice is integral to both the research process and outcome. The project documents its development through logs, video recordings, voice notes, and screen captures, offering insight into decision-making and critical milestones. This documentation serves as a valuable resource for understanding the complexities of blending artistic and technological methodologies in VR-based research.

3.1 Artistic Considerations

Directing the viewer's gaze is an important part of the dramaturgical process in Paul Wenninger's work. It is about the choreographic relationship between gaze, subject and space, with which emotions and content are generated. Giving the viewer the task of directing the gaze, of deciding where the gaze goes, fundamentally changes dramaturgical thinking. Wenninger states in [33]:

"One possibility is to direct the gaze through certain actions. But this misses the fundamental question: What is dramaturgy in VR? A key point is time. It is no longer linear, as in a movie, with a beginning and an end through which the viewer is led, but time is a place, an area, a duration. In VR, it is a place where viewers can linger. They can set something in motion, but they can also distance themselves from it. Loops can allow viewers to return to the same place in time over and over again, but the experience can be different each time. This raises the question: How are spaces constructed? Is orientation in space a question of the design of objects? Is it a question



Figure 1: 35 *Pixel 8a* by Google used in the capturing setup.

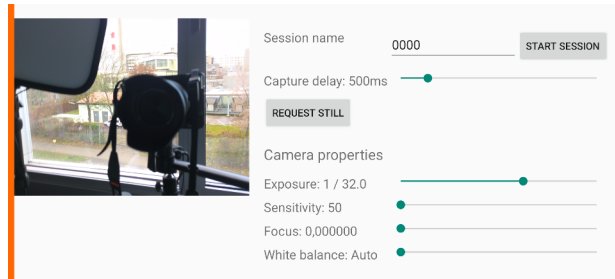


Figure 2: Screenshot of the gui of the camera-control app, based on [1] and extended for the projects needs.

of ethnicity or gender, of sexual orientation? What does orientation have to do with time?

The potential of VR is to be self-oriented. Another important aspect is the position of the user in relation to the place and the events. The position of the viewer can be determined or the action space can be limited. How do we view the events, e.g. in what relationship do the users stand to the object being viewed? How much control do we have over this position? Time, orientation and position are the essential aspects of a dramaturgical concept from which we planned this VR installation.”

3.2 Dramaturgical Aspects

The limited production time of one week for the pixilation recording with the performer Raul Maia, led the team to create a form of loop in which the users are partly guided and partly can decide for themselves how to move. It consists of two indoor situations with the performer in very different actions. The two spaces are connected, on the one hand, by an acoustic event that catapults the viewers into the second space via an outdoor space and, on the other hand, the first space is set up as a miniature model in the second one; as soon as the viewers look inside the model, they are teleported back into the first space.

Having two different scenes loop together, allowed to extend the duration of the resulting VR experience. In addition, Paul Wenninger emphasized that the two scenes should be linked with different experiences for the audience:

- Users should experience the first scene as a fly, following a guided flight path: The path includes stops and close-ups to highlight specific objects or interactions within the scene. Users should retain 6 degrees of freedom (6DOF), allowing them to rotate, look around, and change position for detailed views from different angles.



Figure 3: The capturing setup used in *ExperiMotion 4*.

- In Scene 2, there should be a life-size exploration: In contrast to the guided experience of Scene 1, Scene 2 should allow users to freely explore the environment at life size, creating a sense of agency and discovery. Users can move freely and discover details at their own pace.
- A teleportation portal will connect the two scenes, providing a functional narrative bridge between the path of the fly and the life-size exploration. The entire experience will be designed to loop, allowing users to revisit both scenes multiple times. The looping structure encourages users to notice new details and gain deeper insights with each iteration.

3.3 Animation Capture Setup

Capturing stop-motion animation requires efficient workflows that allow the animator to setup the scene, actors, and props, handle the captured data, and allow for immediate review of the recent frames. While for classic stop motion, a single camera is used, volumetric stop motion works with multiple cameras, triggered synchronously to capture the entire environment for 3D reconstruction. The setup for volumetric capture is built with a volume of about 3x3x3 meters. For practical and economic reasons, the team decided to use 35 smartphones (*Google Pixel 8A*) (see figure 1) with a custom, self-developed camera app, initially based on the work of [1]. This can be remotely controlled via WIFI to achieve synchronized camera setup, triggering and photo downloading (see figure 2). The cameras were arranged in a cylinder-like setup with a distance of approximately 2 meters to the actor (see figure 3).

A custom camera management software, developed in VVVV [10] and deployed on a PC manages all cameras and also serves as a host for a wireless *Xbox* controller, which is used by the director as a trigger for taking synchronised photos. Furthermore, this software also interfaces with an instance of *Dragonframe* [25] running on another PC, which itself is host for a DSLR camera that takes photos at the same time as the smartphones. These photos can be immediately reviewed in *Dragonframe* and serve as a visual reference during a capture session, as well as after a capture break for continuity.

3.4 Gaussian Splatting and Implementation in VR

A single volumetric frame in 3DGS format is created from the single image inputs of all cameras. To create such a frame, the team used *Nerfstudio* with *COLMAP* for camera alignment and *Nerfacto* to train a single 3DGS frame. The captured scenes need to be cleaned to remove non-relevant parts and spatio-temporally



Figure 4: Isolated 3D Gaussian Splatting representation of the performance of the actor.

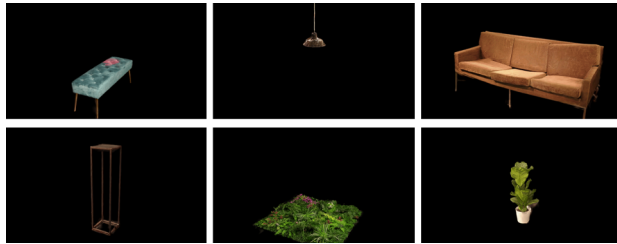


Figure 5: Some of the captured and cleaned assets that were captured and reconstructed to set up the miniature rooms for the VR environment.

aligned. In addition to the main acting stage (see figure 4), single props (e.g. room furniture 5) and background environments were captured individually and then composed into a full 3D scene (see figures 6 and 7).

For initial experiments in VR, *ExperiMotion 4* established a preliminary workflow for rendering Gaussian splats in *Unity3D* using [4]. For VR prototyping, the project allows for loading, cropping, and rendering 3DGS scenes. A sequencer that preloads 3DGS files into VRAM and plays them back in a desired animation sequence was implemented for animation. The system supports a playback rate of 24 splats per second as requested by guest artist Paul Wenninger to suit his creative vision. While this workflow is functional, it is intended as an intermediate solution, due to compositing limitations of the used 3DGS backend, as well as restrictions of available VRAM which cannot store the approximately 3 minutes of animated frames. The ultimate objective is to migrate the entire process to VVVV for improved flexibility in editing and playback.

4 DISCUSSION AND CONCLUSION

While stop-motion animation with puppets or objects has been incorporated into various VR experiences, the integration of pixilation is

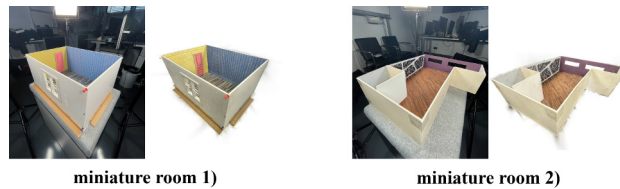


Figure 6: A side-by-side comparison of images of both miniature rooms and their respective 3D Gaussian splatting representations.



Figure 7: A mockup of the initial scene. Note the absence of a ceiling and the external environment, which still need to be reconstructed and integrated.

still a novel approach. Because it is difficult for actors to pose during the recording process, it is impossible to work with one camera that automatically moves around the actor and takes multiple pictures for a volumetric capture. Therefore, the *VRinMotion* team built a spherical setup with 35 smartphones that were triggered simultaneously. A traditional *Dragonframe* setup was used as additional reference for the animator and the performer. The workflow provided an optimal setup for the animation process, enabling the team to record approximately three minutes of pixilation over five days. Guest artist Paul Wenninger, found it intriguing that in VR, framing the protagonist within a traditional (filmic) composition was no longer required. On the one hand, this made the recording process more manageable, on the other hand, the filmmaker was mourning the possibility of guiding the viewer's gaze in VR like he would in a more traditional film project. The setup also did not allow recording the pixilation with any background, because the camera setup had to be removed during post-production. The approach enables the integration of varied scales by capturing background environments or objects in miniature, while human actors and objects are recorded at full size, as seen in traditional pixilation. However, the technique demands significant VRAM capacity to store and render scenes, as evidenced by the challenges faced in playing back approximately three minutes of animated frames in the current workflow (see section 3.4). This memory-intensive nature makes it less practical for longer animations or projects with more extensive sequences.

While the 3D reconstruction of the pixilated actor highlights the haptic quality of the recording, Gaussian splatting deviates from photorealism and offers an artistic alternative to traditional volumetric capture techniques. The unique aesthetic is characterized by the inherent presence of noise and enhances the sense of the fragmented movement of pixilation. It also provides a foundation for further manipulation. In the future, there is potential to influence more aspects of the Gaussian splats' appearance, adding another layer of creative control. By incorporating noise and allowing for dynamic interaction through brush strokes or adjustments to the splatting parameters, this method could provide artists with a flexible, interactive medium for shaping texture, depth, and atmosphere.

The dramaturgical development of *ExperiMotion 4* was driven by creative and technical challenges, particularly the tight production timeline for recording the pixilated animation. A loop structure was developed, linking two distinct indoor scenes featuring the performer Raul Maia that were interconnected through an acoustic event and a visual connection. While the first scene offers a guided fly-through with stops for detailed exploration, the second scene provides users with the freedom to explore at life-size scale. The looping structure not only extends the duration of the experience, but also encourages repeated engagement, as users can revisit the scenes to uncover new

details and deepen their understanding of the virtual environment. For future work, the dramaturgical framework could be extended in various aspects. This includes incorporating interactive elements, allowing users to manipulate objects, change lighting, or trigger different animations. The looping structure could be enhanced with more scenarios to create a more dynamic narrative structure. Further development of spatial audio design, responsiveness to user movement, and improved scene transitions are needed.

While *ExperiMotion 4* is still a project in progress, it already shows the potential to generate novel visual aesthetics in VR. Ultimately, Gaussian splatting opens up new possibilities for digital expression, blending technical innovation with artistic freedom to produce visually compelling and immersive experiences.

ACKNOWLEDGMENTS

The research project *VRinMotion* [AR669] is fully funded by the Austrian Science Fund (FWF).

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